

Operator Logic

New improvement beta 0.1, based on the operational logic and mathematical architecture of the official 2.4

Notes

- Layer correspondence:** the L0/L1/L2 of the retention inference is explicitly mapped onto the three-layer structure—
 - L0 $\triangleq$ Layer Zero (the deepest substrate):** passes only the polarity $\sigma = \{-1 : +1\}$, and enters no integral;
 - L1 $\triangleq$ Layer One (the secondary substrate):** the rule set $\{C, M, S\}$, providing only structural constraints, not participating in integrals as a field;
 - L2 $\triangleq$ Layer Two (the phenomenal layer):** the sole legitimate domain of real mathematics; all continuity equations, integrals, and Hodge decompositions operate at this layer.
- New theorem:** a new operator mathematics is to be built on the full official operator 2.4; this document is its logical draft.
- Glossary:** entries such as hidden retention, transverse circulation, net polarity Π , observable projection $\mathcal{O}$, and S-parity are added.
- Special note:** the “axioms” under this logic apply only to the mathematical analysis conducted under this logic; they are not the ZFC standard of a mathematical system.

Overview: The Three-Layer Structure

Layer	Name	Role	Key Objects	Layer
-------	------	------	-------------	-------

				Mapping
Layer Zero	Deepest substrate	Body (origin)	Great Conservation, ultimate reference frame R^∞	L0
Layer One	Secondary substrate	Function (operating rules)	Non-conserving substrate, conserving substrate	L1
Layer Two	Phenomenal layer	Manifestation (observable)	Incision, discrete quantity $\mathcal{D}$, dual-frame projection $z = a + bi$	L2

Master logical thread:

Great Conservation (continuous) $\rightarrow$ (incision) $\rightarrow \mathcal{D}$ (discrete) $\rightarrow$ (dual-frame projection) $\rightarrow$ observed value z

Every observable phenomenon above the phenomenal layer (the arrow of time, constant temperature, the zeta zeros, hidden retention, etc.) is the manifestation of this thread in a concrete situation — that is, an **example**, not a premise of the system.

Layer Zero: The Deepest Substrate (L0)

Axiom I: Great Conservation (absolutely continuous background)

$$\int_{-\infty}^{+\infty} \square \, dx = 0$$

- **Identity:** the origin of everything; the “nothing” out of which something arises.
- **Properties:** all-inclusive, structureless, unobservable, non-discrete.

- **Logical identity:** the ultimate reference frame R^∞ .
- **Polarity:** inflow (return) is +1; outflow (manifestation) is -1.

Lo Supplement: the absolute layer passes only polarity

- Lo itself **enters no integral**; it passes only the polarity $\sigma = \{-1 : +1\}$, i.e. the counting of “manifestation / return”.
- Any “debt” settled to the absolute layer is judged solely by whether the net polarity is zero.

Layer One: The Secondary Substrate (L1)

Note: Great Conservation is the “body”; these two substrates are the “functions”. They differentiate out of Great Conservation and constitute the lowest-level rules of actual operation.

Axiom II: The Non-Conserving Substrate (dynamical closed loop)

$$0 \leq 1 \geq 0$$

- **Identity:** the logical closed loop of manifestation and return. It is not a “flow” but “the structure that permits flow to exist”.
- **Properties:**
 - **1:** the full state — the extreme of the incision opening, the maximal possibility of manifestation, the unique apex vertex of the structure.
 - **0:** the empty state — the extreme of the incision closing, the endpoint of return.

- **Structural symmetry (key semantics):**
 - **Global symmetry:** the two paths $0 \rightarrow 1$ and $1 \rightarrow 0$ share the same vertex 1; every flow, whatever its direction, must pass through 1 to complete the loop.
 - **Numerical asymmetry:** $0 \rightarrow 1$ is increasing, $1 \rightarrow 0$ is decreasing; step size, direction, and phase may all differ.
 - **Conclusion:** what is symmetric is the structure, not the numbers. This distinction is the basis on which the whole complex structure stands (see Axiom V).
- **The closed loop:** flow is forced to go “from 0 to 1, then from 1 to 0”.
 - This is the structural basis of **all process-type phenomena** (the arrow of time, etc.), not the phenomena themselves (see §4 for examples).
 - It is also the structural basis of **oscillation**.
- **Role:** it defines the boundary of the possibility of “non-conservation”. Without this loop, flow has no channel.

Axiom III: The Conserving Substrate (the metrical rule)

Conserving substrate = Absolute invariant I + Step metric Δ

- **Identity:** the logical premise of observation and measurement.
- **The absolute invariant I :** the anchor under a certain class of rules (e.g. e under the rules of calculus, or the full state 1 under the rules of the incision).
 - It is the zero point of the reference frame, and the identity element of addition:
 $I \oplus x = x$.
 - **Key point:** it is not a number but the standard that defines “invariance”.
- **The step metric Δ :** the rule that defines “how one step is taken” (e.g. linear step

+1, or exponential step $\times e$).

- It provides “order”, giving the throughput of the discrete quantity $\mathcal{D}$ a before-and-after sequence — this is the source of “order” of process-type phenomena (such as the arrow of time) at the phenomenal layer (examples in §4).
- **Role:** it converts the qualitative loop ($0 \leq 1 \geq 0$) of the non-conserving substrate into a quantitative, computable sequence.

L1 Supplement: the rule set {C, M, S}

- **C:** the closed-loop substrate $0 \leq 1 \geq 0$, constraining phenomenal-layer paths to be alternating trajectories between 0 and 1.
- **M:** the conserving substrate (the metrical rule, $I + \Delta$).
- **S:** the frame-swap operator (a structurally symmetric operation; see Axiom V and §6).
- The rule set provides only **structural constraints**; it does not participate in integrals as a field (this is the premise for hidden retention to hold; see §6).

Layer Two: The Phenomenal Layer (L2)

Axiom IV: The incision action and the discrete quantity $\mathcal{D}$

- **The incision:** the dynamical process defined by the non-conserving substrate $0 \leq 1 \geq 0$.
- **$\mathcal{D}$ (the discrete quantity):** the smallest indivisible unit of action that the incision condenses out of the continuous background in the opening–closing process.
- **Logic:**

Great Conservation (continuous) $\xrightarrow{\text{incision}}$ $\mathcal{D}$ (discrete)

Axiom V: The dual-frame projection (the complex structure)

The observed value $z = a + bi$ is the reading of **the same $\mathcal{D}$** in two reference frames:

- **a (real part):** the reading of $\mathcal{D}$ in R_∞ (Great Conservation).
- **b (imaginary part):** the reading of $\mathcal{D}$ in R_1 (the local frame of the incision).
- **i: the frame-rotation operator** — switching the reading from the R_∞ perspective to the R_1 perspective, and the geometric embodiment of the closed loop of the non-conserving substrate.

Why a complex number rather than two real numbers (key semantics):

- The two frames are **structurally isomorphic** (both obey the same $0 \leq 1 \geq 0$ substrate) $\rightarrow$ their readings can be placed in **the same number**;
- but the readings of the two frames are **numerically incommensurable** (R_∞ is the Great-Conservation scale, R_1 is the incision-local scale) $\rightarrow$ the difference must be carried by an **orthogonal direction**.

This is precisely the projection of Axiom II “structural symmetry, numerical asymmetry” onto the phenomenal layer: **structural isomorphism $\rightarrow$ the same number; numerical difference $\rightarrow$ the orthogonal direction (i).**

The structural meaning of $i^2 = -1$: switching perspective twice returns to the origin but flips the polarity — precisely the geometric embodiment of the closed loop $0 \rightarrow 1 \rightarrow 0$ of the non-conserving substrate: go around once, return to the start, and the sign flips.

The projection rule (polarity opposition):

Flow direction	R_∞ reading	R_1 reading	Complex number

Manifestation flow	-1 (outflow)	+1 (inflow)	-1 + i
Return flow	+1 (inflow)	-1 (outflow)	+1 - i

L2 Supplement: the legitimate domain of real mathematics

- L2 is the **sole legitimate domain of real mathematics**: all continuity equations, integrals, and Hodge decompositions operate at this layer.
- We adopt the complex field $z = a + bi$:
 - a : the directly observable degree of manifestation;
 - b : the accompanying phase.

3. The Dynamical Mechanism

1. Complete description of the loop

$$\begin{array}{c}
 \text{Great Conservation} \xrightarrow{\text{outflow}(-1)} \text{Non-conserving substrate } (0 \rightarrow 1) \\
 \xrightarrow{\Delta} \mathcal{D} \text{ manifestation} \\
 \xrightarrow{\text{inflow}(+1)} \text{Great Conservation}
 \end{array}$$

- The **non-conserving substrate** provides the channel ($0 \leq 1 \geq 0$).
- The **conserving substrate** provides the metric (Δ) and the reference zero (I).
- $\mathcal{D}$ is the smallest unit flowing through the channel.

This loop is the system's **only mechanism**. Every observable form at the phenomenal layer is a reading of this loop in a concrete situation (§4, §6).

2. Field equations at the phenomenal layer (within L2)

Continuity equations:

$$\partial_t a + \nabla \cdot \dot{j}_a = s_a$$

$$\partial_t b + \nabla \cdot \dot{j}_b = s_b$$

Steady-state condition:

$$\partial_t z = 0 \implies \partial_t a = 0, \partial_t b = 0$$

Corresponding time-independent equations:

$$\nabla \cdot \dot{j}_a = s_a, \quad \nabla \cdot \dot{j}_b = s_b$$

4. Phenomenal Examples of the System (the layer of examples)

The following three are **of equal standing**: each is a concrete embodiment of the system at the phenomenal layer, not a premise of the system's definition.

They share a single structure — the non-conserving substrate (the channel) $\times$ the conserving substrate (order and metric).

Example 1: The arrow of time

Special note: this is only a logical analysis, used to deconstruct logic; it is absolutely not an argument about time and cannot be used as a proof!

- **Phenomenon:** the one-directionality of process — events have before and after, and cannot flow backwards.
- **System interpretation:** the non-conserving substrate forces the one-way closed loop $0 \rightarrow 1 \rightarrow 0$; the Δ of the conserving substrate assigns an “order” to the throughput of $\mathcal{D}$; acting together, they present the arrow of time at the phenomenal layer.
- **Positioning:** this is the system’s embodiment on “process/order”-type phenomena. The system itself does not define time; it only provides the structure through which time can appear.

Example 2: Constant temperature (22 °C)

Special note: this is only a logical analysis, used to deconstruct logic; it is absolutely not an argument about temperature and cannot be used as a proof!

- **Phenomenon:** energy keeps flowing in, yet the temperature reading stays constant.
- **System interpretation:** the Δ of the conserving substrate limits the generation rate of $\mathcal{D}$, and the 1 of the non-conserving substrate limits the capacity of the incision; when the generation rate reaches the capacity ceiling and is dynamically balanced against the return rate, the R_1 reading b is constant — energy flows through continuously, but the reading does not change.
- **Positioning:** this is the system’s embodiment on “dynamic equilibrium”-type phenomena. Constant temperature is not a conclusion of the system but a reading presented by the loop mechanism under a particular equilibrium condition.

Example 3: The zeta function and the critical line

Special note: this is only a logical analysis, used to deconstruct logic; it is absolutely not an argument for the Riemann hypothesis and cannot be used as a proof!

ζ cascade = the superposition of the dual-frame readings of all $\mathcal{D}$

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s}, \quad s = \sigma + it$$

$$n^{-s} = e^{-\sigma \ln n} \cdot e^{-it \ln n}$$

- $e^{-\sigma \ln n}$: the **intensity reading** of the n -th $\mathcal{D}$ in R_{∞} (the real-part contribution);
- $e^{-it \ln n}$: the **phase reading** of the n -th $\mathcal{D}$ in R_1 (the imaginary-part contribution).

The cascade is precisely the term-by-term superposition, on the complex plane, of the dual-frame readings of all $\mathcal{D}$.

The semantics of convergence = structural return

- When $\sigma > 1$, the R_{∞} reading of each $\mathcal{D}$ decays exponentially, and all condensed units return completely to Great Conservation through the non-conserving substrate, **with no permanent residue**.
- Convergence is not the numerical fact that “the series has a limit” but the structural fact that **the closed path is unobstructed**.

The semantics of the zeros = structural symmetric annihilation

- $\zeta(\rho)=0$: at this complex point, the R_{∞} and R_1 readings of all $\mathcal{D}$ **cancel each other completely** in the complex structure.

- It is not a static return to zero but: after flowing through the leak point, the readings of the two perspectives annihilate each other in a fully symmetric way — return is complete and the residue is zero.

The functional equation = frame swap

$$\zeta(s) = 2^s \pi^{s-1} \sin(\pi s/2) \Gamma(1-s) \zeta(1-s)$$

- $s \leftrightarrow 1-s$ is exactly the interchange of the roles of R_∞ and R_1 (the real-part reading σ and its complementary reading $1-\sigma$ are exchanged, and the sign of the imaginary part flips).
- Because the substrate is **globally symmetric in structure**, this frame swap is a legitimate **structurally symmetric operation**.

The critical line $\sigma=1/2$ = the structural fixed point of the frame swap

- The structural fixed point of the frame swap must satisfy: $\sigma = 1-\sigma \Rightarrow \sigma = 1/2$.
- **It is not “the numerical midpoint of 0 and 1”**: the imaginary part t may take any real value, and the reading itself is not symmetric.
- It is a direct embodiment of the structural symmetry of the non-conserving substrate: **the unique structural position at which the reading remains self-consistent after the frames are swapped**.

The equal standing of the three examples

Example	Phenomenal presentation	Structural source in the system
Arrow of time	process one-directional, with before and after	$0 \rightarrow 1 \rightarrow 0$ closed loop $\times \Delta$'s order
Constant temperature	continuous inflow, constant	balance of generation rate and

(22 °C)	reading	return rate
Critical line	zero resonance concentrated at $\sigma=1/2$	structural fixed point of the frame swap

All three are **readings of the loop mechanism in concrete situations**. The system promises no specific phenomenon; it promises only this: whatever is a phenomenon is a projection of this loop.

5. The Self-Referential Paradox of Metalogic — the System’s Growth Point

Where the paradox lies

- The incision is defined as “the dynamical process defined by the non-conserving substrate $0 \leq 1 \geq 0$ ” (Axiom IV);
- but the non-conserving and conserving substrates themselves are “differentiated” out of Great Conservation;
- and the act of “differentiating” is itself already a minimal incision.

Hence self-reference arises:

To define the incision, one first needs the non-conserving/conserving substrates; to produce the substrates, one first needs the incision.

Deeper still: Great Conservation is the all-inclusive “nothing” and is unobservable. Everything that can be defined or spoken of (including the substrates themselves) is already in a state of manifestation. Therefore, the act of “differentiating the substrates out of Great Conservation” cannot be retroactively defined within the system; it can

only be acknowledged as a self-referential axiom.

Why this is not a defect but a driver

- **Self-bootstrapping:** this paradox frees the system from any external prime mover. The incision is the “first act”; it cannot be proved by itself in the ordinary way, but it supplies the initial tension that keeps the system running.
- **Growth:** precisely because the incision cannot be fully defined, it can keep “incising again” at higher levels — each interpretation of the paradox is a fresh manifestation, so the system loops and grows on itself.
- **Isomorphism with Axiom II:** the non-conserving substrate is itself self-referential — it is both “the structure that permits flow to exist” and can be observed only by means of flow (manifestation/return).

Positioning

This paradox is not a loophole of the system but the starting mechanism of the system as a “metalogic”. It makes the whole architecture logically closed and open: **closed** in that it needs no external premise, **open** in that the tension of the paradox forever produces new computable layers.

6. Hidden Retention

6.1 Definition of hidden retention

A hidden retention h is an L2 steady-state field satisfying the following conditions:

$$\partial_t h = \mathbf{o}, \quad \mathcal{O}h = \mathbf{o}, \quad \Pi(h) = \mathbf{o}$$

where:

- $\mathcal{O}$ is the observable projection, e.g. taking the real-part projection $\mathcal{O}(z) = \text{Re}(z)$;
- $\Pi(\mathbf{h})$ is the net polarity contribution of $\mathbf{h}$, i.e. the net count of “unpaired manifestations” minus “returns”;
- $\Pi(\mathbf{h}) = \mathbf{o}$ means: even if it is a retention, it owes nothing to the absolute layer.

6.2 Two mathematical forms of hidden retention

Form one: transverse circulation

Apply a Hodge decomposition to the steady-state continuity equation:

$$j_a = \nabla\varphi + \nabla\times\psi, \quad \nabla^2\varphi = s_a$$

where:

- $\nabla\varphi$ is the source/sink-driven longitudinal flow; it directly changes a and is therefore the **observable part**;
- $\nabla\times\psi$ is the divergence-free circulation, satisfying $\nabla\cdot(\nabla\times\psi) = \mathbf{o}$; it does not change a and is therefore the **hidden part**.

Thus the first candidate form of hidden retention is:

$$h = j_{\perp} = \nabla\times\psi$$

It is a genuinely non-zero flow within L_2 , yet its existence is completely invisible to the observation of $\rho = a$.

Form two: imaginary-part phase

The second candidate form comes from the imaginary part of the complex field:

$$h = i b$$

If b is steady-state and its sources and sinks are net-balanced, then the distribution of b does not enter the observable real-part channel. Under the frame-swap operator S it flips sign:

$$S: b \mapsto -b$$

Therefore it is an **S-parity** component, with no counterpart in the polarity count of the absolute layer.

6.3 Lemma: balance of entrance and exit of a closed orbit

Lemma (balance of entrance and exit of a closed orbit)

Let γ be a closed trajectory containing:

$$m = \#\{0 \rightarrow 1 \text{ arcs}\}, \quad n = \#\{1 \rightarrow 0 \text{ arcs}\}$$

Since γ is closed:

- at vertex 0 , the number of departures from 0 must equal the number of arrivals at 0 ;
- the only arcs leaving 0 are the $0 \rightarrow 1$ arcs;
- the only arcs entering 0 are the $1 \rightarrow 0$ arcs.

Hence

$$m = n$$

Therefore the polarity contribution of the whole closed orbit is:

$$\Pi(\gamma) = m(-1) + n(+1) = n - m = 0$$

Corollary: under the L1 closed-loop constraint, every closed streamline of any transverse-circulation hidden retention $h = \nabla \times \psi$ carries equal numbers of manifestation and return arcs, and hence contributes zero to the polarity count.

6.4 Theorem: hidden retention is compatible with Great Conservation

Theorem (the balanced structure of hidden retention)

Suppose L2 is in a steady state and the L1 closed loop and the S projection are in effect. If h is any hidden retention, then it must be of one of the following two forms (or their superposition):

$$h = \nabla \times \psi \quad \text{or} \quad h = i b$$

and it satisfies:

$$\Pi(h) = 0$$

Proof:

1. If h is invisible to observables, it cannot contain the longitudinal flow $\nabla \phi$, since the longitudinal flow directly changes a . Hence h can consist only of a divergence-free

circulation and an imaginary part.

2. For the circulation part, by the balance of entrance and exit of a closed orbit (§6.3):

$$\Pi(h_{\perp}) = 0$$

3. For the imaginary part, the polarity count is defined as the net change of the real-part manifestation:

$$\Pi = \int_{\Omega} (\partial_t a + \nabla \cdot j_a) dV$$

The imaginary part b does not enter this functional, and therefore:

$$\Pi(i b) = 0$$

4. By the L2 steady state and the L1 closed-loop constraint, neither of the above two parts produces an “unpaired manifestation”, and hence neither owes anything to Lo.

The theorem holds. ■

6.5 Conclusion: hidden retention is not a permanent residue

- **Hidden retention exists:** it may be a transverse circulation, or it may be an imaginary-part phase;
- **It does not violate Great Conservation:** because $\Pi(h) = 0$;
- **It is not a permanent residue:**
 - the transverse circulation is a balanced cycle of continuous return;
 - the imaginary-part phase is a pure phase freedom with no polarity counterpart;

- **It is compatible with “convergence = structural return: no permanent residue”**: because “no residue” means no unpaired-polarity residue, not the absence of all hidden degrees of freedom.

The true identity of hidden retention is: **a storage channel for paradox tension**. It exists in the form of S-parity or divergence-free circulation, keeping the game tension continuously distributed at the phenomenal layer while incurring no debt to be settled at the absolute layer.

邵昆 (shaokun)

Email: 1947910734@qq.com

<https://orcid.org/0009-0006-0673-5448>

This document is the logical foundation of the system, for internal reference and further iteration.